

Passive Reconnaissance & Google Dorking Guide

Kali Linux Techniques, OSINT Tools, & Execution Troubleshooting

1. Introduction to Passive Reconnaissance

Passive Reconnaissance (or passive OSINT) is the process of gathering intelligence about a target system or organization without directly sending traffic to their servers or triggering intrusion detection systems (IDS). By leveraging publicly accessible databases, DNS resolvers, and third-party search indexes, you collect essential technical context safely.

2. Key Passive Reconnaissance Tools

A. WHOIS (Domain Registration & Metadata)

`whois` queries regional domain registrars to retrieve domain administrative details, ownership metadata, and registration dates.

- **What it reveals:** Domain registrar, registration/expiration dates, authoritative name servers, and administrative contact emails (if not protected by WHOIS privacy).

```
# Run WHOIS query in Kali Terminal
whois targetdomain.com
```

B. Nslookup (DNS Record Enumeration)

`nslookup` queries DNS servers to inspect server-to-IP mappings and specific service configuration records.

- **A/AAAA Records:** IPv4 and IPv6 target IP addresses.
- **MX Records:** Mail exchange servers handling corporate email.
- **TXT Records:** SPF, DMARC policies, and verification tokens.

```
# Query IPv4 address (A Record) using Cloudflare resolver
nslookup -type=A targetdomain.com 1.1.1.1

# Query Mail Exchange (MX) records
nslookup -type=MX targetdomain.com

# Query TXT records for SPF/DMARC validation
nslookup -type=TXT targetdomain.com
```

C. Shodan (Internet-Connected Device Search)

Shodan continuously scans the entire internet to index open ports, service banners, operating systems, and web frameworks. Because Shodan performs the scans independently, searching its database is completely passive for the investigator.

How to get a Free Shodan API Key:

- 1. **Standard Free Account:** Register at `shodan.io` . After verifying your email, go to `account.shodan.io` to view your free community API key.
- 2. **Free Academic Upgrade:** Register using a university email (e.g., `.edu` or `.ac.uk`) to automatically unlock 100 query credits/month, 100 scan credits, Shodan Maps, and image searches.

```
# Initialize Shodan CLI in Kali
shodan init YOUR_FREE_API_KEY

# Query host details by IP
shodan host 8.8.8.8

# Search by domain or organization
shodan search hostname:"targetdomain.com"
shodan search org:"Target Organization"
```

3. Google Dorking Explained

Google Dorking utilizes advanced search operators to filter through Google's index to locate sensitive documents, exposed credentials, unlinked admin panels, and directory listings.

Operator	Description	Example Query
<code>site:</code>	Restricts search results to a specific domain.	<code>site:example.com</code>
<code>filetype:</code> / <code>ext:</code>	Filters by specific document extension.	<code>filetype:pdf</code>
<code>inurl:</code>	Finds specific strings inside the URL path.	<code>inurl:admin</code>
<code>intitle:</code>	Searches for text within the webpage title tag.	<code>intitle:"index of /"</code>
<code>intext:</code>	Searches for specific keywords in the page body.	<code>intext:"password"</code>

Practical Google Dork Examples

```
# 1. Discover Exposed File Directory Listings (Open Indexes)
site:targetdomain.com intitle:"index of /" "parent directory"

# 2. Uncover Admin Logins & Management Dashboards
site:targetdomain.com inurl:admin OR inurl:login OR inurl:portal

# 3. Locate Exposed Credentials or Secrets Files
site:targetdomain.com filetype:env OR filetype:log OR filetype:sql "DB_PASSWORD"

# 4. Map Subdomains (Excluding Main WWW Site)
site:*.targetdomain.com -www.targetdomain.com
```

4. Troubleshooting Terminal Errors: "command not found"

Common Mistake in Linux Terminal

Running a dork like `site:google.com` directly into the Kali Linux shell results in:

```
bash: site:google.com: command not found
```

Why This Happens

Search operators like `site:`, `inurl:`, and `filetype:` are **Google search engine parameters**, not Linux binary commands installed on your operating system.

Where & How to Run Google Dorks Properly

- **In the Web Browser (Recommended):** Open Firefox/Chrome in Kali, go to `google.com`, and paste your query string into the search bar.
- **From Terminal using `googler`:**

```
# Install command-line Google search tool
sudo apt update && sudo apt install googler -y

# Execute Google Dork directly from terminal
googler "site:targetdomain.com filetype:pdf"
```

- **Automated Dork Scrapers:** Use dedicated OSINT tools like `pagodo` to query the Exploit Database Google Hacking Database (GHDB) via proxies.

Summary Checklist

- ✓ Use `whois`, `nslookup`, and `shodan` in the Linux Terminal.
- ✓ Use Google Dorks in a Web Browser or via CLI wrapper tools like `googler`.
- ✓ Combine passive techniques before engaging in active scanning (e.g., Nmap).